INCH-POUND

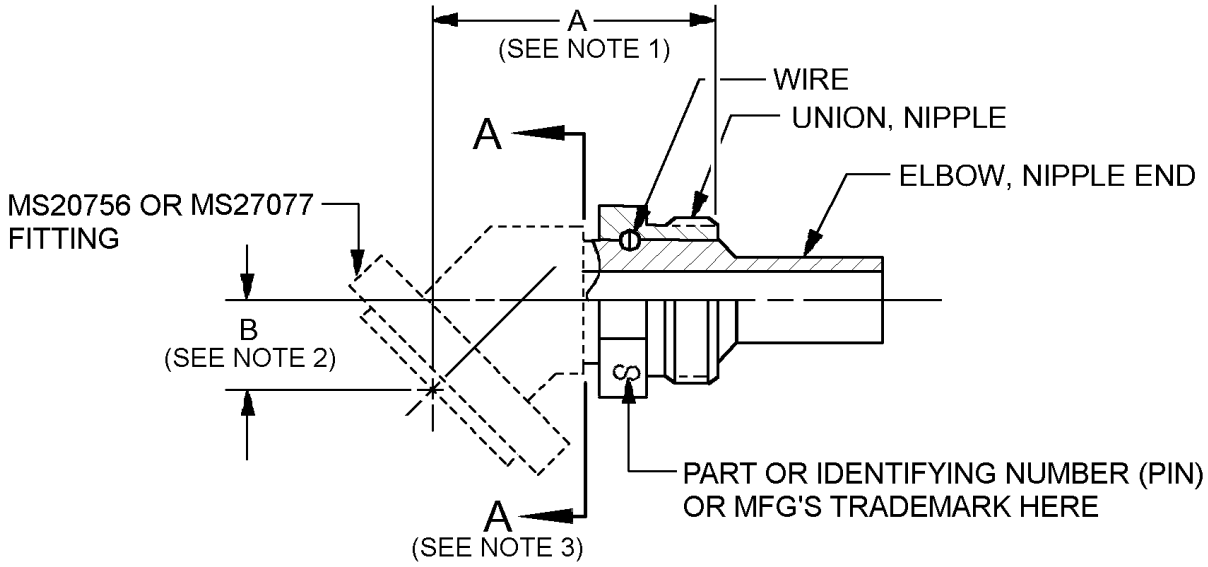
MS27064E
14 January 2014
SUPERSEDING
MS27064D
24 September 2003

DETAIL SPECIFICATION SHEET

ELBOW SUBASSEMBLY, SWIVEL FLANGE, 45°

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-27272.



NOTES:
1. Dimension A, see table II, is measured between the hose end of the threaded section of the union and the free end of the insert at its centerline.
2. Dimension B, see table II, is measured between the centerline of the nipple and the free end of the insert at its centerline.
3. Any bent tube design of subassembly components to the left of plane A-A is acceptable provided that the requirements of this specification sheet and the procurement specification are met.

FIGURE 1. Elbow subassembly illustration.

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REQUIREMENTS

Elbow subassembly illustration. See figure 1

Intended use. This part is a component of MS27056.

Materials. Material and material codes see table I. See applicable specification sheet and MIL-DTL-27272 for material of other components identified in table II for their material requirements.

TABLE I. Materials and material codes.

Material code	Material
Blank	Combination of aluminum alloy and corrosion-resistant steel (CRES)
C	CRES

Finish: Finishes are performed at component level.

Corrosion-resistant steel, do not passivate.

Aluminum alloy. Anodize in accordance with MIL-A-8625, type II, dye blue.

TABLE II. Elbow subassembly composition. 1/ 2/

Size and material code	Wire size and material code	Union, nipple size and material code	Elbow, nipple end size and material code	A 5/ ±.035 (0.89) inch (mm)	B 5/ ±.035 (0.89) inch (mm)
	MS27072 3/	MS27071 3/ 4/	MS27084 3/ 4/		
-8C	-8C	-8C	-8C	1.266 (32.16)	.338 (8.59)
-8			-8D		
-10C	-10C	-10C	-10C	1.324 (33.63)	.375 (9.53)
-10		-10D	-10D		
-12C	-12C	-12C	-12C	1.782 (45.26)	.468 (11.89)
-12		-12D	-12D		
-16C	-16C	-16C	-16C	1.838 (46.69)	.505 (12.83)
-16		-16D	-16D		
-20C	-20C	-20C	-20C	2.052 (52.12)	.569 (14.45)
-20		-20D	-20D		
-24C	-24C	-24C	-24C	2.318 (58.88)	.624 (15.85)
-24		-24D	-24D		

1/ Dimensions are in inches.
2/ Metric equivalents are given for general information only.
3/ MS27071, MS27072, and MS27084 shall be dry film lubricated in accordance with table III, dry film is applied in accordance with MS27071, MS27072, or MS27084 as applicable.
4/ Material code “D” aluminum alloy.
5/ See figure 1.

MS27071, MS27072, and MS27084 parts shall be coated with the same type of dry film lubricant as indicated in the PIN, no mixing of dry film lubricants shall be used on an assembly unless it is of the types specified by the “blank” designator.

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NOTE: Avoid using graphite dry film lubes with aluminum nipples because in a wet environment, graphite becomes corrosive to the aluminum.

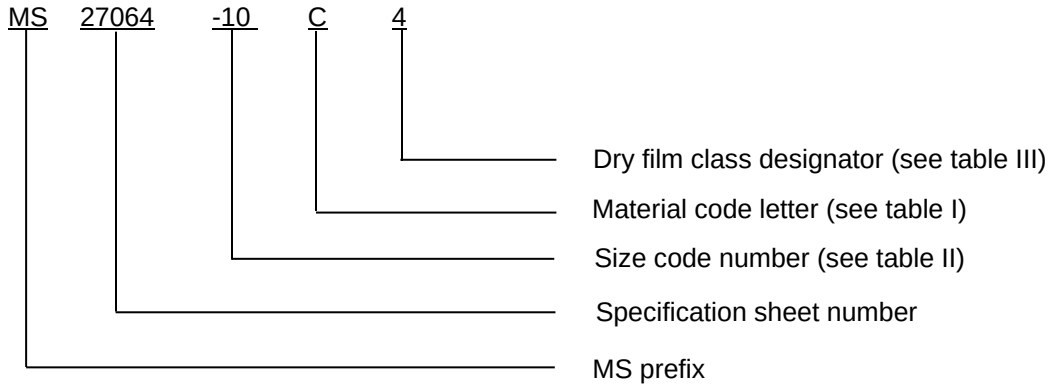
TABLE III. Dry film designator.

Dry film designator	SAE class or type designator	Dry film characteristics
Blank	Any SAE class or type below	N/A
SAE-AS1701	SAE-AS1701 class	SAE-AS1701 temperature ranges °F (°C)
4	4	-65° to +1400°F (-54° to 760°C)
5	5	-65° to +850°F (-54° to 454°C)
6	6	-375° to +850°F (-226° to 454°C)
SAE-AS5272	SAE-AS5272 type	SAE-AS5272 temperature ranges. °F (°C)
7	Type I	-90° to 400°F (-68 to 204°C) endurance life of 250 min minimum
8	Type II	-90° to 400°F (-68° to 204°C) endurance life of 450 min minimum
9	Type III	Color 1 - Natural product color -90° to 400°F (-68 to 204°C) low volatile organic compound with an endurance life of 450 min minimum
10	Type III	Color 2 - Black color -90° to 400°F (-68 to 204°C) low volatile organic compound with an endurance life of 450 min minimum
Dry film designator	MIL classification	Dry film characteristics
MIL-PRF-46010 1/	---	MIL-PRF-46010 temperature ranges. °F (°C)
11	N/A	Color 1 natural product color, -90° to 400°F (-68 to 204°C) solvent resisting
12	N/A	Color 2 - Black color -90° to 400°F (-68 to 204°C) solvent resisting

1/ Not for aerospace usage.

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Part or Identifying Number (PIN): The PIN consists of prefix “MS”, the specification sheet number, dash number for 45° elbow subassembly size, letter for material, and a blank or number for dry film lubricant. Unassigned PIN’s shall not be used.



PIN examples:

MS27064-10	indicates a 45° elbow subassembly 10, a combination of aluminum and CRES, and with dry film class designator “blank”.
MS27064-10C	indicates a 45° elbow subassembly size 10, CRES with dry film class designator “blank”.
MS27064-10C4	indicates a 45° elbow subassembly size 10, CRES with dry film class designator 4.

Identification of product. The PIN and Government Entity (CAGE) Code or manufacturer’s trademark shall be permanently marked on the elbow subassembly.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-27272, this document references the following:

MIL-A-8625	MS27072
MIL-PRF-46010	MS27056
MS20756	MS27084
MS27077	SAE-AS1701
MS27071	SAE-AS5272

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CONCLUDING MATETRIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2013-101)

Review activities:
Army - AR, AT, MI
Navy - MC, SA, SH
Air Force - 71

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